Balanced Vortex Dynamics: The Sawyer–Eliassen Equation

Why eyewall heating produces ascent and eye subsidence

Source of Notes

Roger Smith's Tropical Cyclone notes:

Link To Roger's Notes

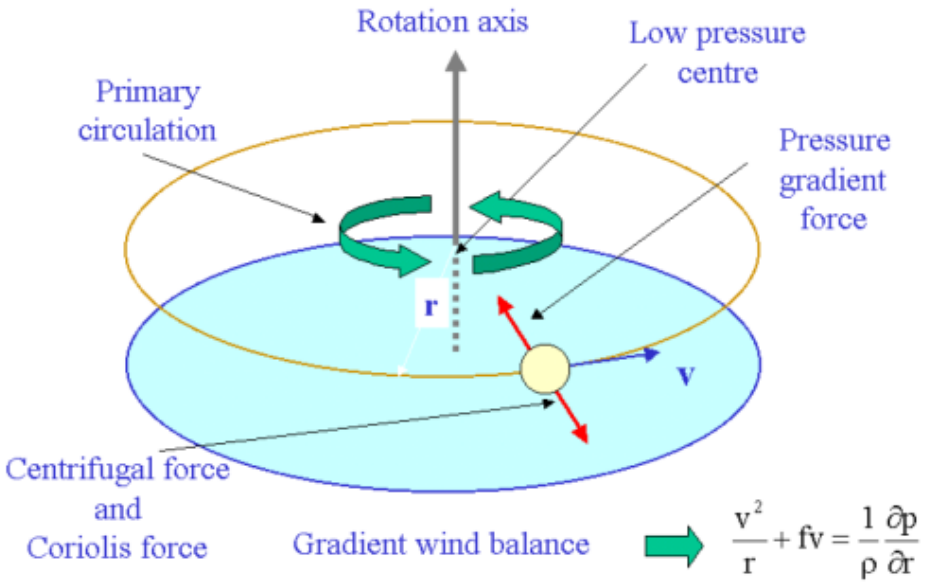
All figures are from these notes

Learning goals

1. Distinguish primary and secondary circulation.
2. Explain how heating and momentum forcing produce a balanced overturning circulation.
3. Interpret inertial stability and static stability as resistance to displacement.
4. Explain compensating descent in the eye without confusing it with a universal local heating response.

Two circulations, one storm

- **Primary:** strong azimuthal wind $v(r, z)$; approximately gradient-wind and hydrostatic balance.
- **Secondary:** weaker radial–vertical flow (u, w): boundary-layer inflow, eyewall ascent, upper outflow and broader subsidence.
- The secondary circulation redistributes angular momentum and heat, allowing the primary vortex to evolve while remaining nearly balanced.



Inward pressure-gradient force balances outward centrifugal and Coriolis terms for cyclonic flow.

$$\frac{1}{\rho}\frac{\partial p}{\partial r} = \frac{v^2}{r} + fv.$$

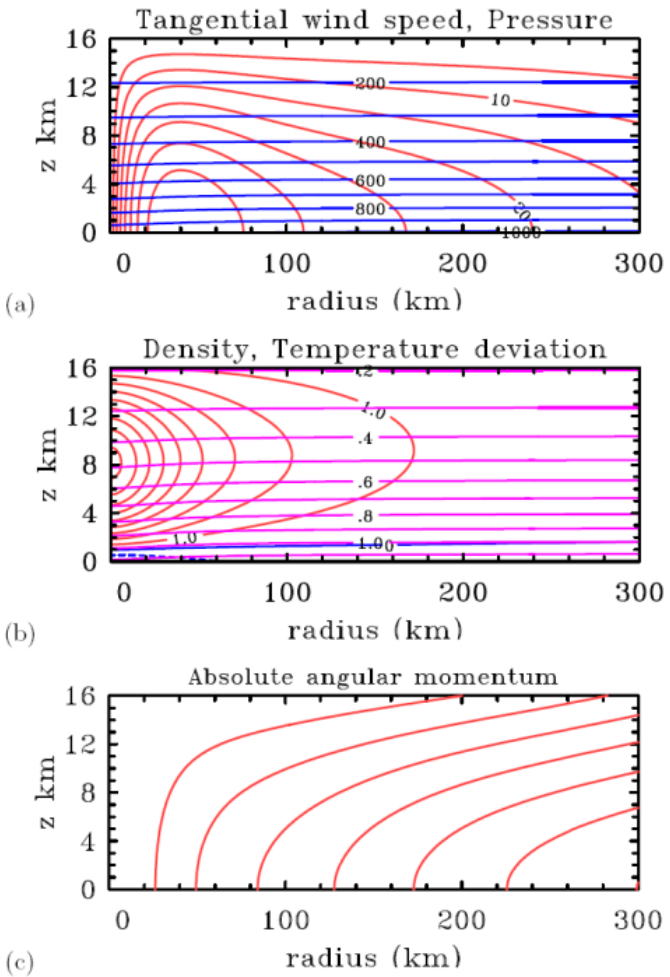
This is a *radial* force balance; it does not say that radial velocity is exactly zero everywhere.

A mathematical overview: what is balanced?

$$\frac{1}{\rho} p_r = \frac{v^2}{r} + fv, \qquad p_z = -\rho g.$$

- Combining these relations constrains radial temperature gradients and vertical shear (thermal-wind compatibility).
- Diabatic heating changes temperature; torque changes angular momentum. Either forcing would disrupt compatibility unless an overturning flow adjusts the fields.
- The Sawyer–Eliassen (SE) equation determines that adjustment circulation.

Structure of a balanced warm-core vortex



(a) Tangential winds and sloping pressure surfaces.

(b) Mid/upper-level warm anomaly and density structure.

(c) Outward-sloping absolute angular-momentum surfaces $M = rv + \frac{1}{2}fr^2$.

The warm core and vertical wind shear are linked by balance, not independent ingredients.

How the secondary circulation is represented

$$\rho r u = -\frac{\partial \psi}{\partial z}, \qquad \rho r w = \frac{\partial \psi}{\partial r}.$$

- $\psi(r, z)$ is a mass streamfunction; its contours trace overturning cells.
- This definition automatically satisfies axisymmetric mass continuity.
- Solve one diagnostic equation for ψ rather than separately predicting u and w.
- The response depends on the *existing* vortex, the location of heating/torque, and boundary conditions.

The Sawyer–Eliassen equation: schematic form

$$\underbrace{\mathcal{L}_{I^2,N^2,B}[\psi]}_{\text{balanced resistance and coupling}} = \underbrace{\mathcal{F}_Q(Q)}_{\text{heat forcing}} + \underbrace{\mathcal{F}_F(F_\lambda)}_{\text{tangential momentum forcing}} .$$

- N^2: resistance to vertical displacement (static stability).
- I^2: resistance to radial displacement (inertial stability).
- B: baroclinic coupling / tilt of the background vortex.
- For a stably balanced, elliptic problem, forcing induces a spatially distributed response; it is *not* confined to the forcing location.

This is a schematic operator, not the fully expanded SE equation; precise coefficients depend on the thermodynamic formulation and streamfunction convention.

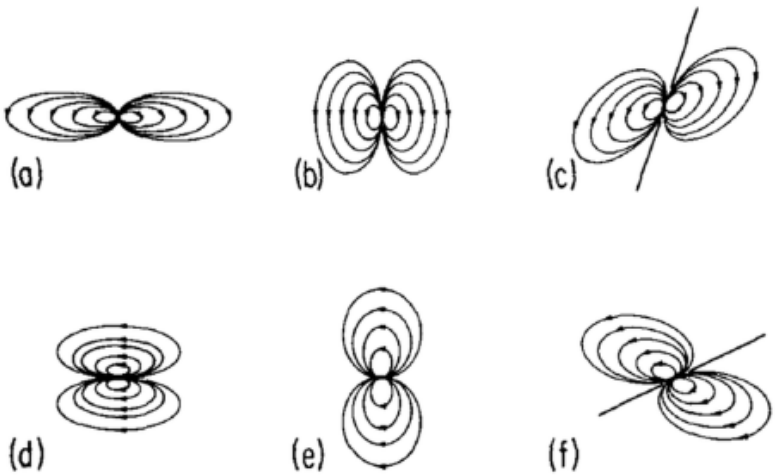
What is inertial stability?

Angular momentum (M) and Inertial Stability (I^2):

$$M = rv + \frac{1}{2} fr^2, \qquad I^2 = \frac{1}{r^3} \frac{\partial M^2}{\partial r}$$

- Large I^2: strong restoring tendency against radial displacement; circulation becomes radially confined and relatively vertically elongated.
- Small I^2: easier radial adjustment; a broader response.
- Near the intense inner core, high inertial stability can help localize the response to eyewall heating.

Idealized responses to point forcing

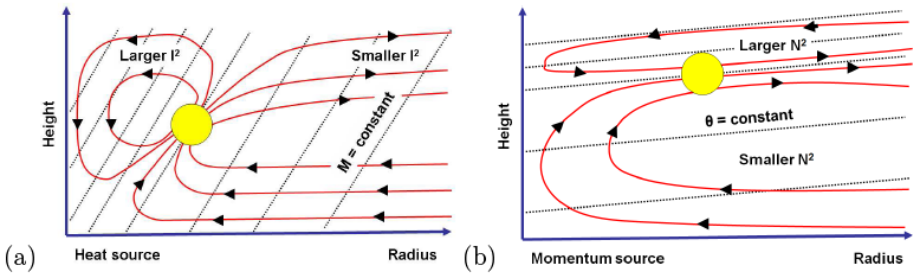


Top row (a–c): heating. Weak versus strong inertial stability changes horizontal extent; baroclinicity tilts the cells.

Bottom row (d–f): cyclonic momentum forcing. Predominantly radial response; static stability shapes its vertical extent.

Each source produces a dipole of overturning cells, not just motion at the source.

Idealized responses to point forcing



(a) Heating: ascent through the heat source, approximately along M surfaces; return flow and compensating descent adjacent to it.

(b) Momentum source: outward flow through the source, approximately along isentropes; compensating return flow.

The strongest flow follows the path of least balanced resistance, not a strictly vertical or horizontal line.

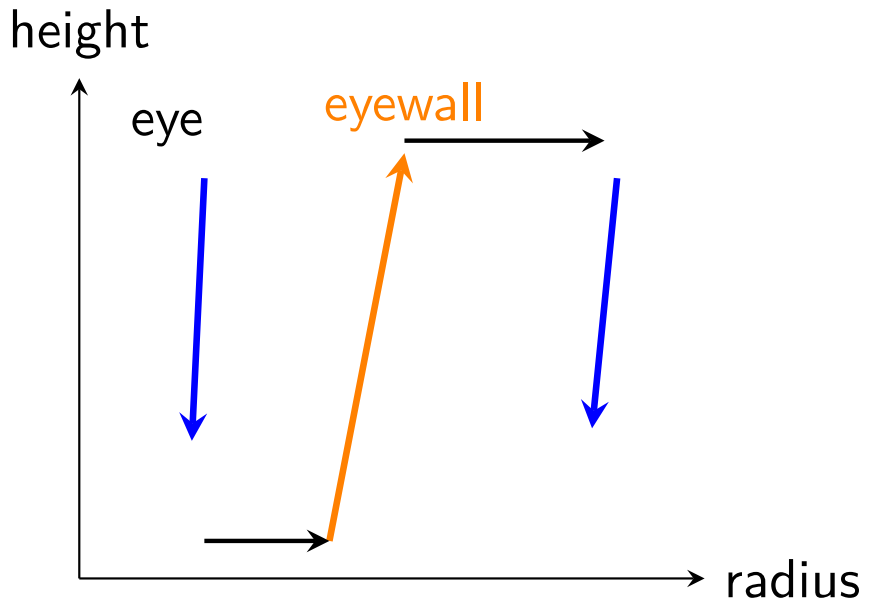
Apply the theory to an eyewall heat source

1. Condensation releases latent heat in a narrow annulus near the radius of maximum wind.
2. A balanced response requires ascent through that heating and return flow outside the updraft.
3. Mass continuity and vortex stability distribute compensating subsidence on both sides of the eyewall; the inner branch can descend into the eye.
4. Subsiding eye air warms adiabatically, contributing to a warm, relatively dry eye.

Important: the partition between inner-eye and outer-region descent depends on vortex structure, heating geometry, boundaries, and unbalanced processes.

Why is the eye warm if it is not heated strongly?

- Eyewall: latent heating and strong ascent.
- Eye: weaker convection, subsidence and compressional warming.
- Gradient-wind/hydrostatic adjustment links warm-core structure to pressure and wind.
- The strongest warm anomaly is commonly aloft; the eye need not have persistent vigorous descent at every height.



A second perspective: evolution of central pressure

- During vortex intensification, adjustment may favor eye subsidence and warming.
- But **balance alone does not require perpetual descent**: in a steady, unforced balanced vortex, no ongoing vertical motion is necessary.
- SE theory explains the circulation *forced by* heating and torque; transient eye formation and mixing may require additional dynamics.

Concept check: three predictions

1. Keep heating fixed, increase inner-core inertial stability: does its radial influence widen or narrow?
2. Move eyewall heating farther from the axis: must eye subsidence remain equally strong?
3. Remove heating but retain a balanced vortex: must air continue to sink in the eye?

Answers: (1) generally narrows; (2) no; (3) no.

Where balance theory is incomplete

- Boundary-layer friction and supergradient flow can violate gradient-wind balance.
- Strong, localized convective bursts and vortex asymmetries are not axisymmetric balanced motions.
- Regions with symmetric instability may make the standard SE operator non-elliptic.
- Eye subsidence is a plausible balanced response, not a claim that all eye descent is caused solely by local eyewall heating.

Summary and source

- The primary vortex supplies the stability and geometry; heating and torque drive the secondary circulation.
- Eyewall heating induces ascent plus a distributed compensating circulation, including possible eye subsidence.
- Eye warming follows compression during descent; persistent descent is not a prerequisite of a steady balanced eye.

Adapted conceptually from R. K. Smith, *Dynamics of Tropical Cyclones*, sections 3.2–3.8, figures 3.2, 3.5, 3.6, 3.7. Original figure URLs: `https://www.meteo.physik.uni-muenchen.de/~roger/TCLecs/`. Figures 3.6 and 3.7 attributed there to Eliassen (1951) and Willoughby (1995), respectively.